ANGLO-CHINESE JUNIOR COLLEGE

JC2 Preliminary Examination

Higher 2

GEOGRAPHY **9173/02**

Paper 2 **28 August 2024**

3 hours

Additional Materials: Insert
Cover Sheet
Answer Paper

READ THESE INSTRUCTIONS FIRST

You should follow the instructions on the cover sheet. If you need additional answer paper, ask the invigilator.

Write your index number and name on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer Question 1 in **Section A.**
Answer Question 2 in **Section B**.
Answer **one** question in **Section C.**

Start your answer to each question on a fresh sheet of answer paper.

At the end of the examination, your answers should be separated and securely fastened into **2 bundles**. Attach a cover sheet on top of bundle 2.

- Bundle 1: Answers to Question 1. *Name and index number to be clearly indicated on first page of answers.*
- Bundle 2: Cover sheet and Answers to Question 2 and Question 3 or 4.

The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **5** printed pages and **1** blank page.

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Answer Questions 1 and 2 and
EITHER Question 3 **OR** Question 4

Section A

Cluster 4 Fieldwork

1 A group of six students worked on an investigation on flood risk around the Punggol Waterway which flows through the Punggol Waterway Park and Punggol residential estate. The Punggol Waterway connects the Serangoon River in the east to the Punggol River in the west. These three bodies of waters are dammed to create the Punggol-Serangoon Reservoir. They selected two sites X and Y along the Punggol Waterway for their investigation.

Their hypothesis is that flood risk is higher in built-up areas relative to vegetated areas.

To assess the flood risk of the Punggol Waterway Park and Punggol residential estate, they collect data on infiltration rates and selected river variables. They collected data on a weekend in November from 9 am to 4 pm. At each site, two students worked on the infiltration test, while four worked on collecting data on selected river variables. All six of them worked at site X, and then then moved onto site Y after lunch.

For infiltration, the students chose a location at each site, as shown in Resource 2. The equipment used included an infiltrometer made of white plastic tubing with a diameter of 10 cm and a height of 30 cm. The infiltrometer was hammered into the ground using a pebble found nearby. A student used a pail to fetch water from the river for the infiltration test. Water was poured to the top of the infiltrometer, and using a stopwatch, the water level dropping in the infiltrometer was read every 5 minutes. When water reached the base of the infiltrometer, water was poured into the infiltrometer again.

For the river variables, the following data were collected:

- Channel width – Two students on the footbridge that span the river at each site used a tape measure to stretch across the bridge to measure the river width from above.
- Channel depth – This was obtained by extending a long nylon string with a plumb bob (weight) from the bridge at regular intervals of 5 metres across the span of the footbridge from one river to the opposite bank. The plumb bob was lowered into the river to touch the river bed, and the river depth was obtained by measuring the length of the string and the plumb bob.
- River velocity – This was measured using a floating orange to travel a 5-metre distance along the river. Velocity readings were taken once at each site.

Resource 1 is a land use map of the area adjacent to the Punggol Waterway, showing sites X and Y on the river. Resource 2 shows the photographs of sites X and Y. Resource 3 shows the infiltration rates at site X and site Y. Resource 4 shows the results of the river investigation.

(a) Explain if the students' hypothesis and choice of sites are appropriate. [4]

(b) With reference to Resources 1 and 2, describe how the students could reduce the impacts of their investigation. [4]

(c) Explain how the students can mitigate the potential risks of their fieldwork investigation at the sites shown in Resource 2. [6]

(d) Explain how the reliability of the infiltration and river data shown in Resources 3 and 4 can be improved. [6]

(e) Explain the data that the students should gather if they were to conduct a supplementary questionnaire survey on flood risk in the Punggol area. [5]

(f) Explain the extent of the usefulness of the river data shown in Resource 4 in the investigation of flood risk. [5]

(g) The student concluded that vegetated areas show lower flood risk compared to built-up areas. With reference to Resources 1, 2, 3 and 4, evaluate the validity of the students' conclusion. [10]

Section B

Cluster 3 Sustainable Future and Climate Change

2 Resource 5 shows the distribution of the Small Island Developing States (SIDS). Resource 6 shows the temperature anomaly versus sea level rise. Resource 7 shows the impacts of climate change on aquatic and terrestrial ecosystems. Resource 8 shows an infographic about the impacts of climate change on SIDS.

(a) Using Resource 5, describe the locations of the SIDS in the world. [4]

(b) Explain the role of human activities in accelerating warming through positive feedbacks. [4]

(c) Using Resource 6, describe the relationship between temperature anomaly and sea level rise. [4]

(d) Using Resources 6 and 7, explain the impacts of climate change on aquatic and terrestrial ecosystems. [6]

(e) Explain why there are variations in negative impacts of contemporary climate change due to uneven changes in temperature and precipitation across places. [6]

(f) Using Resource 8, explain how countries can respond to contemporary climate change. [6]

Section C

Answer EITHER question 3 OR question 4.

3 Evaluate the strategies in creating liveable cities for different social groups. [20]

4 'Mitigation strategies are the best response to contemporary climate change.'

Evaluate this statement. [20]

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